3.10 Technique 7: Solution of Non-Linear Equations Reducible to Linear Form

Here we shall discuss three classical non-linear differential equations which can be reduced to linear form and can, accordingly, be solved. They include:

a. Bernoulli's Equation,

b. Ricatti's Equation,

c. Clairaut's Equation.

3.11 Bernoulli's Equation

Sometimes we come across differential equations of the form,

$$\mathbf{y}' + P(x)\, y = Q(x)\, \mathbf{y}^{\mathbf{n}}, \qquad (3.\ 11\ 1)$$

where n is an integer. This equation is known as ***Bernoulli's equation***, after **Jacob Bernoulli**, or, **Bernoulli *of Bale*** (1654–1705), who studied it in 1695. On dividing this equation by $\mathbf{y}^{\mathbf{n}}$ and multiplying by (– n + 1), it can be reduced to linear form as follows,

$$(-n+1)\ \mathbf{y}^{-\mathbf{n}}\ \mathbf{y}' + (-n+1)\, P\ \mathbf{y}^{-\mathbf{n}+\mathbf{1}} = (-n+1)\, Q.$$

On substituting $v = \mathbf{y}^{-\mathbf{n}+\mathbf{1}}$, so that $v' = (-n+1)\, y^{-n}\ y'$, this equation becomes,

$$\frac{\mathbf{dv}}{\mathbf{dx}} + (1-n)\, P\, v = (1-n)\, Q, \qquad (3.\ 11\ .\ 2).$$

which is linear in v, and can be solved by Technique 6, illustrated in the preceding section.

Example 3.11.1

Solve $y' + (\frac{1}{x})y = x^2\ \mathbf{y}^{\mathbf{6}}$.

Solution

On dividing the given equation by $\mathbf{y}^{\mathbf{6}}$, and multiplying by (– 6 + 1) = – 5 , we obtain,

$$-5\,\mathbf{y}^{-\mathbf{6}}\ y' - 5\,(\mathbf{y}^{-5})\,/\,x = -5x^2.$$

Putting $\mathbf{v} = \mathbf{y}^{-\mathbf{5}}$ (and thus, $\mathbf{v}' = -5\mathbf{y}^{-\mathbf{6}}\mathbf{y}'$) in this equation we get,

$$\mathbf{v}' - \mathbf{5}\frac{\mathbf{v}}{\mathbf{x}} = -5\, x^{\mathbf{2}},$$

which is linear differential equation in v. Here, P = – 5/x , and Q = – 5x^2, and so,

$$\text{I. F.} = \mathbf{e}^{\int \mathbf{P\,dx}} = \mathbf{e}^{-5\int (1/x)dx} = \mathbf{e}^{-5\ln x} = \mathbf{e}^{\ln(1/x^5)} = \frac{\mathbf{1}}{\mathbf{x}^5}.$$

Therefore, the formula (3.9.5) yields,

$$\boldsymbol{v = y^{-5} = x^5[\int \frac{-5x^2}{x^5} dx + c] = x^5[-5\int x^{-3}\,dx + c] = x^5[\frac{5}{2x^2} + c]}$$

or,

$$\boldsymbol{y^{-5} = \frac{5}{2}x^3 + c\,x^5},$$

which is the required solution of the given differential equation.

Note. In general an equation of the form,

$\boldsymbol{f'(y)}$ + P f(y) = Q,

(where P and Q are functions of x only), on substituting v for f(y) becomes,

dv/dx + P v = Q,

which is linear in v, and can be solved easily.

Examples 3.11.2

Solve $\mathbf{y}'$ + xy / (1 – x^2) = $xy^{1/2}$

Solution

Dividing the given equation by $y^{1/2}$, and multiplying by (– ½ +1) = ½, it becomes,

$$\frac{1}{2}y^{-1/2}\frac{dy}{dx} + \frac{1}{2}\left(\frac{x}{1-x^2}\right)y^{1/2} = \frac{1}{2}x\,.$$

We put $\boldsymbol{v = y^{1/2}}$ (and so, $\frac{1}{2}y^{-1/2}\frac{dy}{dx} = \frac{dv}{dx}$) in this equation to get,

$$\frac{dv}{dx} + \frac{x}{2(1-x^2)}v = \frac{x}{2}\,.$$

This equation is linear in $\boldsymbol{v}$. Here,

$$\text{I.F.} = \boldsymbol{e^{\int P\,dx} = e^{\int\{x/2(1-x^2)\}dx}} = \boldsymbol{e^{\frac{-1}{4}\ln(1-x^2)}} = (1-x^2)^{-1/4}.$$

Thus, using the formula given by Eqn (3.9.5) we obtain,

$$\boldsymbol{v = (1-x^2)^{1/4}[\int (1-x^2)^{-1/4}(x/2)dx + c]}$$

$$= \boldsymbol{(1-x^2)^{1/4}[-\frac{(1-x^2)^{3/4}}{3} + c]},$$

or, $\boldsymbol{y^{1/2} = -\frac{1-x^2}{3} + c(1-x^2)^{1/4}}$,

which is the required solution.

Example 3.11.3

Solve : x $y - y' = y^3 e^{-x^2}$

Solution

In order to free the right hand side from y^3, we divide the given equation by y^3, and multiply by (– 3 + 1) = – 2, to obtain,

$$2y^{-3}y' - 2y^{-2}x = -2e^{-x^2}.$$

We put $v = y^{-2}$, so that, $v' = -2y^{-3}y'$. Thus the above equation becomes,

$$v' + 2vx = 2e^{-x^2}.$$

This equation is linear in v. Here, P = 2x, and Q =2e^{-x^2}. Therefore,

$$\text{I.F.} = e^{\int P\,dx} = e^{\int 2x\,dx} = e^{x^2}.$$

Thus, using the formula given by Eqn (3.9.5), we obtain,

$$v = e^{-x^2}[\int e^{x^2}\, 2e^{-x^2}\, dx + c]$$

$$= y^{-2} = e^{-x^2}\,[2x + c].$$

Or,

$$y = \pm \frac{e^{(x^2/2)}}{\sqrt{2x + c}},$$

which is the required solution of the given differential equation.

EXERCISE – 3. 7

(Bernoulli's Equations)

Note: Answers are given against the respective questions within square brackets.

Solve the following differential equations,

1. $\mathbf{y}' + (2/x)\, y = x^2\, y^{-2}$ $[y^3 = x^3/3 + c/x^6]$
2. $\mathbf{y}' + xy = xy^5$ $[1 = y^4 + cy^4\, \mathbf{e^{2x^2}}]$
3. $\mathbf{y}' + y = xy^3$; $[1/y^2 = x + 1/2 + c\, e^{2x}]$
4. $\mathbf{y}' + y/x = (x^3 - 3)\, y^2$, $x \neq o$; $[1/y - 3x \ln|x| + x^4/3 - cx = 0]$
5. $\mathbf{y}' = x\, y^2 + xy$; $[1/y + 1 - c\, \mathbf{e^{-x^2/2}} = 0]$
6. $dy + (1/x)\, y\, dx = 3\, x^2\, y^2\, dx$; $[xy\, (c - 3x^2/2) = 1]$
7. $dx + (2/y)\, x\, dy = 2\, x^2\, y^2\, dy$; $[x^{-1}\, y^{-2} = c - 2y]$
8. $2\,\mathbf{y}' - y/x = 5\, x^3\, y^3$; $[x\, y^{-2} + x^5 = c]$
9. $3\,\mathbf{y}' + 3\, y/x = 2\, x^4\, y^4$; $[x^{-3}\, y^{-3} + x^2 = c]$
10. $\mathbf{y}' \cos x - y \sin x = y^2 \cos^2 x$; $[y\,(c - x) = \sec x]$
11. $y - \mathbf{y}' \cos x = y^2 \cos x\, (1 - \sin x)$; $[\frac{1}{y} = \frac{\cos x}{1+\sin x}(\sin x + c)]$
12. $2\, x^3\, \mathbf{y}' = y\, (y^2 + 3\, x^2)$; $[y^2\, (c - x) = x^3]$
13. $x\, \mathbf{y}' + y = y^2 \ln x$; $[1/y = cx + 1 + \ln x]$
14. $\mathbf{y}' = y\, (x\, y^3 - 1)$ $[y^{-3} = x + 1/3 + c\, e^{3x}]$

By substitution reduce to linear form and solve:

16. $dx - 2\, xy\, dy = 6\, x^3\, y^2\, \mathbf{e^{-2y^2}}\, dy$; $[x^{-2}\, \mathbf{e^{2y^2}} = c - 4\, y^3]$
17. $dy + y\, dx = 2\, x\, y^2\, e^x\, dx$; $[1 = y\, e^x\, (c - x^2)]$
18. $\mathbf{y}' = y - x\, y^3\, e^{-2x}$ $[e^{2x} = y^2\, (x^2 + c)]$
18. $x\, dy - \{y + x\, y^3\, (1 + \ln x\}\, dx = 0$ $[\ \mathbf{\frac{x^2}{y^2} = -\frac{2}{3}x^3(\frac{2}{3} + \ln x) + c}\ ]$

Solve the initial value problems,

19. $x^2\, \mathbf{y}' = y\, (x + y^2)$, $y(0) = 1$; $[x^2 + 2\, xy = 0]$
20. $\mathbf{y}' + y/2x = x/y^3$, $y(1) = 2$; $[x^2\, y^4 = x^4 + 15]$
21. $(12\, e^{2x}\, y^2 - y)\, dx = dy$, $y(0) = 1$; $[y^{-1}\, e^{-x} = 13 - 12\, e^x]$
22. $3\, y^2\, \mathbf{y}' + y^3/(x + 1) - 8\,(x + 1) = 0$, $y(0) = 0$; $[y^3\, (x + 1) = \mathbf{\frac{8}{3}}\, \{(x + 1)^3 - 1\}]$
23. $x\, dy + 2y\, dx = (x^8 y^{-2}/3)\{3y(x^{-2})^2 + 2y\, x^{-2}\}dx$, $y(1) = 0$;
 $[y^2 = (1/x^4)\, \{x^8/4 + 2x^{10}/15 - 23/120]$

3.12 Ricatti's Equation

Ricatti's equation is of the form,

$$y' = P(x) + Q(x)\,y + R(x)\,y^2 . \qquad (3.12.1)$$

This is linear if R(x) = 0, and non-linear otherwise. The study of this equation is attributed to the Italian count, mathematician and philosopher **Jacopo Francesco Ricatti (1676 -1754)**. It is one of the simplest differential equation of 1st order that cannot, in general, be integrated by **quadratures.** This fact was proved by French mathematician **Joseph Liouville (1809 -1882)**, in 1841.

When the solution of a differential equation is expressed by a formula involving one or more integrations, it is said that the equation is solvable by **quadrature**, and the formula is called a **closed-form solution**. Historically, the term **quadrature** has its origin in the problem of finding the area of a plane figure, by means of **integration**. Therefore, in plane geometry, a problem of quadrature, is a problem about the area of a plane figure.

Every differential equation is not solvable analytically. Furthermore, not all analytically solvable differential equations can be solved by quadrature. For example, except in some special cases, the Ricatti's equation cannot be solved by quadrature.

Usually we seek a general solution of the Ricatti's equation (3.12.1), which should contain one arbitrary constant. But, depending upon the terms, P(x), Q(x) and R(x), there may be no apparent way of obtaining such a solution. However, if we can somehow (often by observation, guessing, or trial-and-error) manage to find one particular solution, $y = y_1(x)$, say, we can obtain the general solution by any of the following two methods:

First Method

We substitute $y = y_1 + \frac{1}{z}$, i.e., $y' = y_1' - \frac{1}{z^2} z'$, in Eqn.(3.12.1) to get,

$$y_1' - \frac{1}{z^2}\, z' = P + Q\left(y_1 + \frac{1}{z}\right) + R\left(y_1 + \frac{1}{z}\right)^2$$

$$= P + Q\,y_1 + \frac{Q}{z} + R\,y_1^2 + \frac{2\,R\,y_1}{z} + \frac{R}{z^2}$$

$$= (P + Q\,y_1 + R\,y_1^2) + \left(\frac{Q}{z} + \frac{2\,R\,y_1}{z} + \frac{R}{z^2}\right).$$

But $y_1' = P + Q\,y_1 + R\,y_1^2$. So, the above equation becomes,

$$-\frac{1}{z^2}\, z' = \frac{Q}{z} + \frac{2\,R\,y_1}{z} + \frac{R}{z^2},$$

i.e.,

$$z' + (Q + 2R\,y_1)\,z = -R, \qquad (3.12.2)$$

which is a linear differential equation in z and can be solved by Technique 6. Thus, the general solution of the given differential equation is given as, $y = y_1 + \frac{1}{z}$, where, z is the solution of Eqn. (3.12.2).

Second Method

We can also solve Eqn.(3.12.1) by substituting $y = y_1 + z$, and thus $y' = y_1' + z'$, in it, where, $y = y_1$, is a previously known solution of this differential equation. Accordingly we obtain,

$$\begin{aligned} y_1' + z' &= P + Q\,(y_1 + z) + R\,(y_1 + z)^2 \\ &= P + Q\,y_1 + Q\,z + R\,y_1^2 + 2\,R\,y_1\,z + R\,z^2 \\ &= (P + Q\,y_1 + R\,y_1^2) + (Q\,z + 2\,R\,y_1\,z + R\,z^2). \end{aligned}$$

But $y_1' = P + Q\,y_1 + R y_1^2$. So the above equation gives,

$$z' - (Q + 2\,R\,y_1)\,z = R\,z^2, \qquad (3.12.3)$$

which is a Bernoulli's equation with $n = 2$, and can be solved by the method discussed above. Therefore, the general solution of the given differential equation is given as, $y = y_1 + z$, where, z is the solution of Eqn. (3.12.3).

Example 3.12.1

Solve

$$x^3\,y' = x^2\,y + y^2 - x^2, \qquad (3.12.4)$$

given that, $y = x$, is a particular solution of this equation.

Solution

The given equation can be written as,

$$y' = -\frac{1}{x} + \frac{1}{x}\,y + \frac{1}{x^3}\,y^2, \qquad (3.12.5)$$

which is a Ricatti's equation with $P = -\frac{1}{x}$, $Q = \frac{1}{x}$, and $R = \frac{1}{x^3}$. Since $y = x$ is a particular solution of this equation, so we substitute,

$$y = x + \frac{1}{z}, \text{ i.e., } y' = 1 - \frac{1}{z^2}\,z'.$$

Thus Eqn.(3.12.5) becomes,

$$\begin{aligned} 1 - \frac{1}{z^2}\,z' &= -\frac{1}{x} + \frac{1}{x}\left(x + \frac{1}{z}\right) + \frac{1}{x^3}\left(x + \frac{1}{z}\right)^2 \\ &= -\frac{1}{x} + 1 + \frac{1}{xz} + \frac{1}{x} + \frac{2}{x^2 z} + \frac{1}{x^3 z^2}. \end{aligned}$$

Thus,

$$-z' = \left(\frac{1}{x} + \frac{2}{x^2}\right) z + \frac{1}{x^3},$$

i.e.,

$$z' + \left(\frac{1}{x} + \frac{2}{x^2}\right) z = -\frac{1}{x^3}. \qquad (3.12.6)$$

It is a linear differential equation in z. Its solution is given by,

$$z = e^{-\int P\,dx}\left[\int e^{\int P\,dx}\, Q\, dx + c\right], \qquad (3.12.7)$$

where, $P = \left(\frac{1}{x} + \frac{2}{x^2}\right)$ and $Q = \frac{-1}{x^3}$. Here,

$$e^{\int P\,dx} = e^{\int (1/x + 2/x^2)\,dx} = e^{\int (1/x)\,dx}\; e^{\int (2/x^2)\,dx} = e^{\ln x}\; e^{-2/x} = x\; e^{-2/x},$$

$$e^{-\int P\,dx} = 1/e^{\int P\,dx} = (1/x)\,e^{2/x}.$$

Therefore, the formula (3.12.7) gives,

$$z = \frac{1}{x}\, e^{2/x}\left[\int x\, e^{-2/x}(-1/x^3)\, dx + c\right]$$

$$= \frac{1}{x}\, e^{2/x}\left[\int (-1/x^2)\, e^{-2/x}\, dx + c\right]$$

We let $v = -2/x$, i.e., $dv = \frac{2}{x^2}\, dx$, so that the above equation gives,

$$z = -\frac{1}{2x} + \frac{c}{x}\, e^{2/x},$$

Thus, the general solution of the given differential equation is,

$$y = x + \frac{1}{z} = x + \frac{1}{\frac{-1}{2x} + \frac{c}{x}\, e^{2/x}}.$$

EXERCISE – 3.8

(Ricatti's Equation)

Solve the following Ricatti's equations for the given y_1:

1. $y' + 2 + y - y^2 = 0, \quad y_1 = 2;$ $\quad [y = 2 + \dfrac{1}{ce^{-3x} - 1/3}]$
2. $y' - 1 + x + y - x\,y^2 = 0, \quad y_1 = 1;$ $\quad [y = 1 + \dfrac{1}{-0.5 + c\,e^{-x^2}}]$
3. $y' + \dfrac{4}{x^2} + \dfrac{y}{x} = y^2, \quad y_1 = \dfrac{2}{x}$ $\quad [y = \frac{2}{x} + \dfrac{1}{cx^{-3} - x/4}]$
4. $y' = 2\,x^2 + \dfrac{y}{x} - 2\,y^2, \quad y_1 = x$
5. $y' = e^{2x} + (1 + 2\,e^{2x})\,y + y^2, \quad y_1 = -\,e^x$ $\quad [y = -\,e^x + \dfrac{1}{cx^{-x} - 1}]$
6. $y' = \sec^2 x - (\tan x)\,y + y^2, \quad y_1 = \tan x;$ $\quad [y = \cos x + \dfrac{1}{\ln|\sec x + \tan x|} + c\,]$
7. $y' = 1 + x^2 - 2xy + y^2, \quad y_1 = x;$ $\quad [y = x + \frac{1}{c - x}]$
8. $y' = -\dfrac{1}{x^2} - \dfrac{y}{x} + y^2, \quad y_1 = \frac{1}{x};$ $\quad [y = \frac{1}{x} + \frac{2x}{(c - x^2)}]$
9. $y' = \dfrac{2\cos^2 x - \sin^2 x + y^2}{2\,\cos\,x}, \quad y_1 = \sin x;$ $\quad [y = \sin x + \dfrac{1}{(c\cos x - \frac{1}{2}\sin x)}]$
10. $y' = x + y(1 - 2x) - y^2\,(1 - x), \quad y_1 = 1;$ $\quad [y = 1 + 1/(x + ce^x)]$
11. $x(x - 1)\,y' - (2x + 1)y + y^2 + 2x = 0, \quad y_1 = x;$ $\quad [y = (x^2 + c)/(x + c)\,]$
12. $y' = x + \{(1/x) - x^3\}\,y + xy^2, \quad y_1 = x^2;$ $\quad [y = x^2 - (x\,e^{x^4/4})/(c + \int x^2\,e^{x^4/4}\,dx)]$
13. $y' = e^{2x} - y\,e^x + e^{-x}\,y^2, \quad y_1 = e^x;$ $\quad [y = e^x - (\,e^{(2x - e^x)})/(c + \int e^{(x - e^x)}\,dx\,]$

Determine a particular solution by inspection and then solve the following Ricatti's equations,

14. $y' = 5 + \dfrac{4}{x}y + \dfrac{1}{4x^2}y^2,$ $\quad [y_1 = -\,2x$
15. $y' = (1 - y)\,(\frac{1}{x} - \frac{1}{10} + \frac{y}{10}),$ $\quad [y_1 = 1$
16. $y' = \frac{6}{x^2} - 2\,y^2,$ $\quad [y_1 = \frac{2}{x}$

17. $y' = -1 + 2y - y^2$, [$y_1 = 1$

18. $y' = 3\,e^{4x} + 2y - 12\,y^2$, [$y_1 = \frac{e^{2x}}{2}$

19. $y' = e^{2x} + (1 + \frac{5}{2}e^{x})y + y^2$, [$y_1 = -2\,e^{x}$

20. $y' = \frac{4}{x^2} - \frac{y}{x} - y^2$, [$y_1 = \frac{2}{x}$

21. $y' = 1+x+2x^2 \cos x - (1+4x \cos x)y+2(\cos x)\,y^2$,

[$y_1 = x$, $y = x - (\sin x - \cos x + ce^{x})^{-1}$]

3.13 Clairaut's Equation

Any differential equation of the form,

$$y = px + f(p), \qquad (3.13.1)$$

where p = dy/dx, and f is any differentiable function of p, is called **Clairaut's Equation** .

Differentiating the above equation w.r.t. x we get (since, $\frac{dy}{dx} = p$),

$$p = p + x\,p' + f'(p)\,p'.$$

Thus,

$$[x + f'(p)]\,\frac{dp}{dx} = 0,$$

which gives,

$$\frac{dp}{dx} = 0, \text{ or} \qquad (3.13.2)$$

$$x + f'(p) = 0. \qquad (3.13.3)$$

Thus the solution of the given differential equation must satisfy one of the following pairs of equations,

$$\left.\begin{aligned} y &= p\,x + f(p) \\ \frac{dp}{dx} &= 0 \end{aligned}\right\}, \qquad (3.13.4)$$

or,

$$\left.\begin{aligned} y &= p\,x + f(p) \\ x &= -f'(p) \end{aligned}\right\} \qquad (3.13.5)$$

Solution of the First Pair of Equations (Eqns. (3.13.4))

The solution of the second equation, $\frac{dp}{dx} = 0$ is given as,

$$p = c \text{ (arbitrary constant).}$$

Therefore, substituting this value of p we obtain the solution of the first equation as,

$$y = cx + f(c), \qquad (3.13.6)$$

which is the general solution, obtainable directly from the given differential equation, by ubstituting c for p in it.

Solution of the Second Pair of Equations (Eqns (3.13.5))

Substituting the value of x from the second equation, i.e.,

$$x = -f'(p) \qquad (3.13.7)$$

into the first equation we obtain,

$$y = f(p) - p\, f'(p). \qquad (3.13.8)$$

If $f(p)$ is neither a linear function of p, nor a constant, both the above equations, i.e., Eqns.(3.13.7) and (3.13.8), are parametric equations of a non-linear solution of the differential equation (3.13.1).

Since Eqn.(3.13.6), the general solution of the given differential equation, represents a straight line for each value of c, the solutions given by Eqns.(3.13.7) and (3.13.8) cannot be a special case of this equation. As such they give a ***singular solution.*** The singular solution may be interpreted geometrically as the ***envelope*** of the family of integral curves.

Example 3.13.1.

Solve the differential equation:

$$y = p\,x + p^3. \qquad (3.13.9)$$

Solution.

The given differential equation is of the form $y = px + f(p)$, which is the Ricatti's equation with $f(p) = p^3$. Therefore, as discussed above, the **general solution** of the given differential equation is given as,

$$y = cx + c^3\,. \qquad (3.13.10)$$

The parametric equations of the solution are,

$$x = -f'(p)\,, \text{ and } \; y = f(p) - p\, f'(p),$$

i.e.,

$$x = -3p^2, \quad \text{and} \quad y = p^3 + p\,(-3p^2) = -2p^3.$$

Eliminating p, these equations give,

$$\frac{x3}{-27} = \frac{y2}{4}\,. \qquad (3.13.11)$$

It gives,

$$4\,x^3 + 27\,y^2 = 0, \qquad (3.13.12)$$

which is a **singular solution** of the given differential equation as it cannot be obtained from the general solution. Therefore, Eqns.(3.13.10) and (3.13.12) give the pair of solutions (i.e., general and singular solutions) of the given differential equation.

EXERCISE-3.9
(Clairaut's Equation)

Note: Answers are given against the respective questions within square brackets.

Solve the following Clairaut's Equations and obtain a singular solution in each case:

1. $y = x y' + 1 - \ln y'$ [$y = c x + 1 - \ln c$; $y = 2 + \ln x$]
2. $y = xy' + (y')^{-2}$ [$y = cx + c^{-2}$;]
3. $y = x y' - (y')^3$ [$y = cx - c^3$; $27 y^2 = 4 x^3$]
4. $y = (x + 4) y' + (y')^2$ [$y = c(x + 4) + c^2$;]
5. $x y' - y = e^{y'}$ [$y = cx - e^c$; $y = x \ln x - x$]
6. $y - x y' = \ln y'$ [$y = cx + \ln c$]
7. $y = x y' - (y')^2$ [$y = cx - c^2$; $y = x^2/4$]
8. $y = x y' + \{1 - (y')^2\}^{1/2}$ [$y = cx + (1 - c^2)^{1/2}$; $y = (1 + x^2)^{1/2}$]
9. $x y' = y - \{1 + (y')^2\}^{1/2}$ [$y = cx + (1 + c^2)^{1/2}$; $y = (1 - x^2)^{1/2}$]
10. $y = x y' - e^{-y'}$ [$y = cx - e^{-c}$; $y = x - x \ln(-x)$]
11. $y = x y' + y' - (y')^2$ [$y = cx + c - c^2$; $y = (x + 1)^2/4$]
12. $y = x y' + \{1 + y'\}^{1/2}$ [$y = c x + (1+c)^{1/2}$; $y = -x - 1/4x, x<0$]
13. $x = y (dx/dy) + \{dx/dy\}^{1/2}$ [$x = cy + c^{1/2}$; $4xy + 1 = 0$]
14. $y = (x + 1) y' - 1$ [$y = cx + c - 1$; $x = -1$]

Review Exercise

Solve the following differential equations:

1. $dy/dt = e^{-y} \sin t$, $y = 0$ at $t = 0$. [$y = ln\,(2 - \cos t)$]

2 $\cos y\, dx + (1+ e^x ce^x) \sin y\, dy = 0$, $y(0) = 0$ [$1+ e^x = 2 \cos y$]

3. $(1+ 2y^2)\, dy = y \cos x\, dx$, $y(0) = 1$ [$ln\,|y| + y^2 = \sin x + 1$]

4. $2(y-1)\, dy = (3x^2+ 4x + 2)\, dx$, $y(0) = -1$; [$y = 1-\sqrt{\mathbf{x^3+2x^2+2x+4}}$)]

5. $(1- x^2)\, y^2 + \mathbf{y}' = 0$, $y(1) = 2$; [$1/y = x - x^3/3 - 1/6$]

6. $\dfrac{dr}{dt} = -rt$, $r(0) = r_0$; [$r = r_0 e^{-t^2/2}$]

7. $\mathbf{y}' = \sec y$, $y(0) = 0$; [$y\ \sin^{-1} x$]

8. $2x\mathbf{y}' = 3y$, $y(1) = 4$; [$y = 4x^{3/2}$ (semicubical parabola)]

9. $\cos x + 3y^2 y' = 0$ [$y^3 + \sin x = c$]

10. $\mathbf{e^{y-x} = y' - x\, e^{y+x}}$ **[$\mathbf{e^{-y} = c + e^{-x} - (x-1)\, e^x}$]**

11. $\mathbf{y(1 + x)\, dx + x(1 + y)\, dy = 0;}$ **[$\mathbf{x + y + \mathit{ln}\,|xy| = c}$]**

12. $\mathbf{cosec\, y\ dx + \sec x\, dy = 0;}$ **[$\mathbf{\sin x - \cos y = c}$]**

13. $y' + \mathbf{y^2 \sin x = 0;}$ **[$\mathbf{1/y + \cos x = c}$]**

14. $(y\, dx + dy)\, e^x = 0$, $y(0) = 2$; [$y = 2e^{-x}$]

15. $(2xy - 3)\, dx + (x^2 + 4y)\, dy = 0$, $y(1) = 2$; [$x^2 y - 3x + 2y^2 = 7$]

16. $(x - y)\, dx + (2y - x)\, dy = 0$, $y(0) = 1$; [$y = \frac{1}{2}\,(x + \sqrt{4-x^2}\,)$]

17. $(2xy^3 + y \cos x)\, dx + (3x^2 y^2 + \sin x)\, dy = 0$; [$x^2 y^3 + y \sin x = c$]

18. $(y \cos x + 2x e^y)\, dx + (\sin x + x^2 e^y - 1)\, dy = 0$; [$y \sin x + x^2 e^y - y = c$]

19. $(ye^x + 2x \cos y)\, dx + (e^x - \cos y - x^2 \sin y)\, dy = 0$; [$ye^x+ x^2 \cos y - \sin y = c$]

20. $(y \sec^2 x + \sec x \tan x)\, dx + (\tan x + 2y)\, dy = 0$; [$y \tan x + \sec x + y^2 = c$]

21. $(y + y \cos xy)\, dx + (x + x \cos xy)\, dy = 0$; [$xy + \sin xy = c$]

22. $(3x^2 y^2 - 2xy^3)\, dx + (2x^3 y - 3x^2 y^2)\, dy = 0$; [$x^3 y^2 - x^2 y^3 = c$]

23. $(6xy + 2y^2 - 5)\, dx + (3x^2 + 4xy - 6)\, dy = 0$; [$3x^2 y + 2y^2 x - 5x - 6y = c$]

24. $(x^2 - x + y)\, dx + (y^2 - 3y + x)\, dy = 0$; [$x^3/3 - x^2/2 + xy + y^3/3 - 3y^2/2 = c$]

25. $(3x^2 + 4xy)\, dx + (2x^2 + 2y)\, dy = 0$; [$x^3 + 2x^2 y + y^2 = c$]

26. $(x^2 - 4xy - 2y^2)\, dx + (y^2 - 4xy - 2x^2)\, dy = 0$; [$x^3 - 6x^2 y + 6xy^2 + y^3 = c$]

27. $(x + 2/y)\, dy + y\, dx = 0$; $[\, xy + \ln y^2 = c\,]$

28. $3x^2 y\, dx + x^3\, dy = 0$; $[\, x^3 y = c\,]$

29. $(2x - y)\, dx + (2y - x)\, dy = 0$; $[\, x^2 - xy + y^2 = c\,]$

30. $(x - 2y)\, dx + (4y - 2x)\, dy = 0$; $[\, x - 2y = c\,]$

31. $(y - x^3)\, dx + (x + y^3)\, dy = 0$; $[\, 4xy - x^4 + y^4 = c\,]$

32. $(x^2 y - 2x)\, dx + (y^2 + \frac{x^3}{3})\, dy = 0$; $[\, x^3 y - 3x^2 + y^3 = c\,]$